Feature Redundancy Clustering
(Within-Sequence Spearman, Top 30 Features)

Merge threshold (|ρ| > 0.85)

Distance (1 − |ρ|)

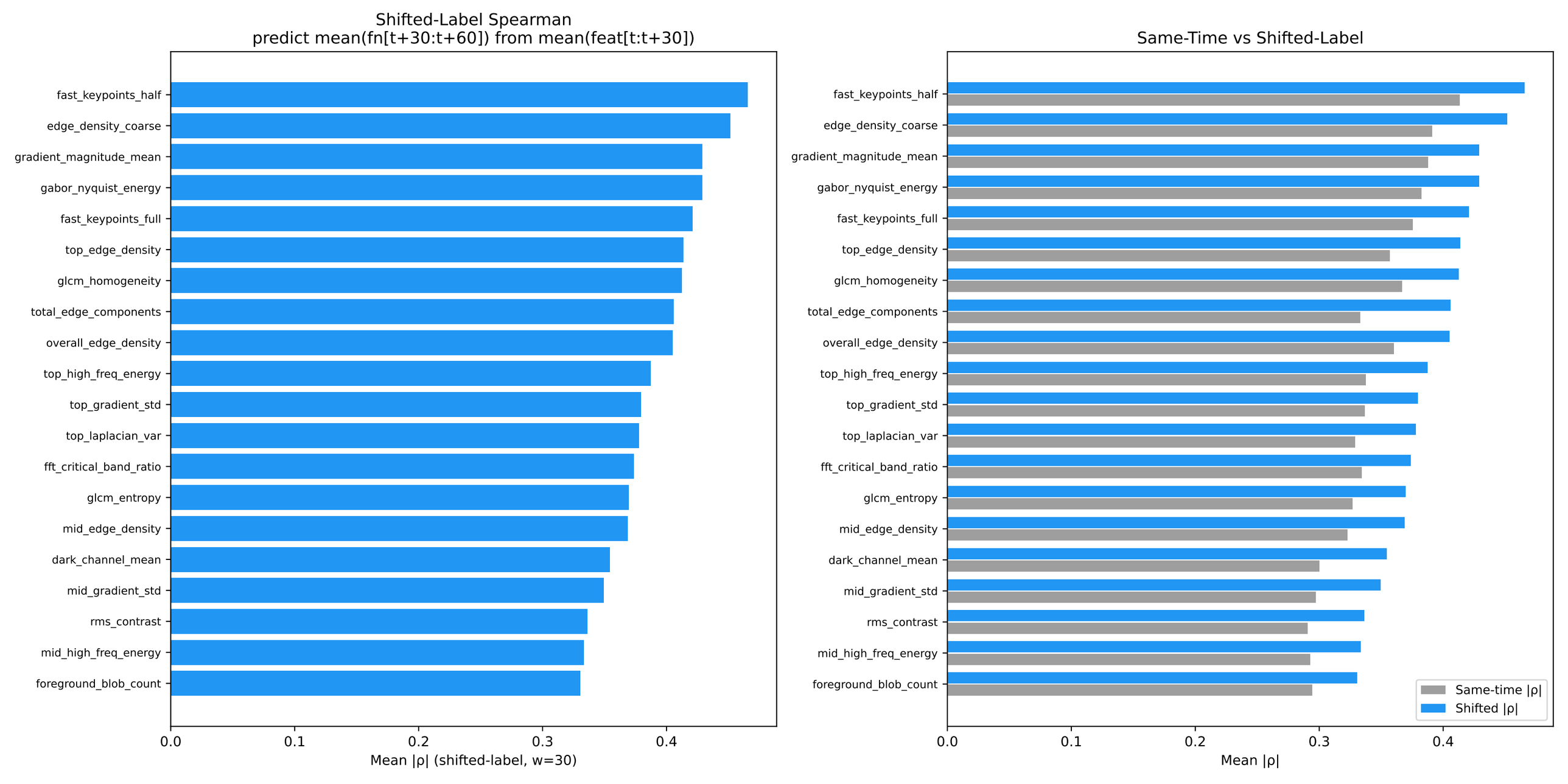
0.0
0.1
0.2
0.3
0.4
0.5
0.6

foreground_blob_count
top_high_freq_energy
top_edge_density
top_gradient_std
top_laplacian_var
dark_channel_mean
fft_critical_band_ratio
fft_high_freq_energy_ratio
mid_edge_density
mid_gradient_std
mid_high_freq_energy
edge_density_coarse
edge_fine_coarse_ratio
rms_contrast
glcm_homogeneity
glcm_entropy
glcm_energy
gabor_nyquist_energy
total_edge_components
small_edge_components
fast_keypoints_half
overall_edge_density
edge_density_fine
glcm_contrast
gradient_magnitude_mean
spatial_frequency
gradient_magnitude_std
fast_keypoints_full
laplacian_mean_abs
laplacian_variance

Within-Seq Feature Correlation (Mean |ρ|)
Blue/Bold = Selected features

fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
glcm_contrast
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
top_edge_density
gradient_magnitude_std
laplacian_mean_abs
top_high_freq_energy
top_gradient_std
fft_critical_band_ratio
total_edge_components
small_edge_components
top_laplacian_var
glcm_entropy
laplacian_variance
mid_edge_density
fft_high_freq_energy_ratio
edge_fine_coarse_ratio
dark_channel_mean
mid_gradient_std
foreground_blob_count
glcm_energy
mid_high_freq_energy
rms_contrast

1.0
0.8
0.6
0.4
0.2
0.0


Shifted-Label Spearman
predict mean(fn[t+30:t+60]) from mean(feat[t:t+30])
fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
gabor_nyquist_energy
fast_keypoints_full
top_edge_density
glcm_homogeneity
total_edge_components
overall_edge_density
top_high_freq_energy
top_gradient_std
top_laplacian_var
fft_critical_band_ratio
glcm_entropy
mid_edge_density
dark_channel_mean
mid_gradient_std
rms_contrast
mid_high_freq_energy
foreground_blob_count
0.0
0.1
0.2
0.3
0.4
Mean |ρ| (shifted-label, w=30)
Same-Time vs Shifted-Label
Same-time |ρ|
Shifted |ρ|
Mean |ρ|

```
======================================================================
FEATURE SELECTION SUMMARY
======================================================================

Input:     65 features, top 30 analyzed for redundancy
Method:    Hierarchical clustering (merge if within-seq |ρ| > 0.85)
Selected:  20 non-redundant features

Validation: Shifted-label Spearman (window = 30 frames = 1 sec)
  Task: predict mean(fn_nano[t+30:t+60]) from mean(feature[t:t+30])

Feature                              Same-time    Shifted Dropped from cluster
-------------------------------------------------------------------------------------------
fast_keypoints_half                     0.4134     0.4656 (singleton)
edge_density_coarse                     0.3910     0.4516 edge_fine_coarse_ratio
gradient_magnitude_mean                 0.3878     0.4290 glcm_contrast, spatial_frequency, gradient_magnitude_std
gabor_nyquist_energy                    0.3824     0.4288 (singleton)
fast_keypoints_full                     0.3753     0.4208 laplacian_mean_abs, laplacian_variance
glcm_homogeneity                        0.3666     0.4124 (singleton)
overall_edge_density                    0.3602     0.4051 edge_density_fine
top_edge_density                        0.3569     0.4136 (singleton)
top_high_freq_energy                    0.3376     0.3873 (singleton)
top_gradient_std                        0.3366     0.3796 (singleton)
fft_critical_band_ratio                 0.3340     0.3738 fft_high_freq_energy_ratio
total_edge_components                   0.3329     0.4059 small_edge_components
top_laplacian_var                       0.3290     0.3780 (singleton)
glcm_entropy                            0.3269     0.3695 glcm_energy
mid_edge_density                        0.3226     0.3686 (singleton)
dark_channel_mean                       0.3000     0.3544 (singleton)
mid_gradient_std                        0.2972     0.3493 (singleton)
foreground_blob_count                   0.2941     0.3305 (singleton)
mid_high_freq_energy                    0.2927     0.3334 (singleton)
rms_contrast                            0.2905     0.3361 (singleton)

======================================================================
COPY-PASTE FOR TRAINING:
======================================================================
SELECTED_FEATURES = [
    "fast_keypoints_half",
    "edge_density_coarse",
    "gradient_magnitude_mean",
    "gabor_nyquist_energy",
    "fast_keypoints_full",
    "glcm_homogeneity",
    "overall_edge_density",
    "top_edge_density",
    "top_high_freq_energy",
    "top_gradient_std",
    "fft_critical_band_ratio",
    "total_edge_components",
    "top_laplacian_var",
    "glcm_entropy",
    "mid_edge_density",
    "dark_channel_mean",
    "mid_gradient_std",
    "foreground_blob_count",
    "mid_high_freq_energy",
    "rms_contrast",
]
```